4.2 Exponential Smoothing: Aggregate-Daily

2026-04-08

Goal

Apply exponential smoothing to the aggregate-daily Rossmann sales series and evaluate its forecasting performance using a holdout set.

We compare a baseline ETS model and an STL+ETS alternative, then evaluate both on the same holdout period using RMSE.

```r
library(tidyverse)
library(forecast)

daily_data <- readr::read_csv(
  "../../data/processed/store_type_a_train_aggregate_daily.csv",
  show_col_types = FALSE
) |>
  dplyr::mutate(
    Date = as.Date(Date)
  ) |>
  dplyr::arrange(Date)

head(daily_data)
```

```
## # A tibble: 6 x 11
##   Date         Sales OpenStores PromoStores PromoShare SchoolHolidayShare
##   <date>       <dbl>      <dbl>       <dbl>      <dbl>              <dbl>
## 1 2013-01-01    2907          1           0          0             1
## 2 2013-01-02 3920386        602           0          0             1
## 3 2013-01-03 3528027        601           0          0             0.939
## 4 2013-01-04 3610215        601           0          0             0.939
## 5 2013-01-05 2964587        600           0          0             0.0664
## 6 2013-01-06    6254          3           0          0             0.0664
## # i 5 more variables: DayOfWeek <dbl>, Year <dbl>, Month <dbl>, IsoWeek <dbl>,
## #   IsWeekend <dbl>
```

```r
set.seed(443)

holdout_prop <- 0.20
n_total <- nrow(daily_data)
n_train <- floor((1 - holdout_prop) * n_total)

train_subset <- daily_data |> dplyr::slice(1:n_train)
holdout_subset <- daily_data |> dplyr::slice((n_train + 1):n_total)
holdout_horizon <- nrow(holdout_subset)

holdout_split <- tibble::tibble(
  n_total = n_total,
  n_train = n_train,
```

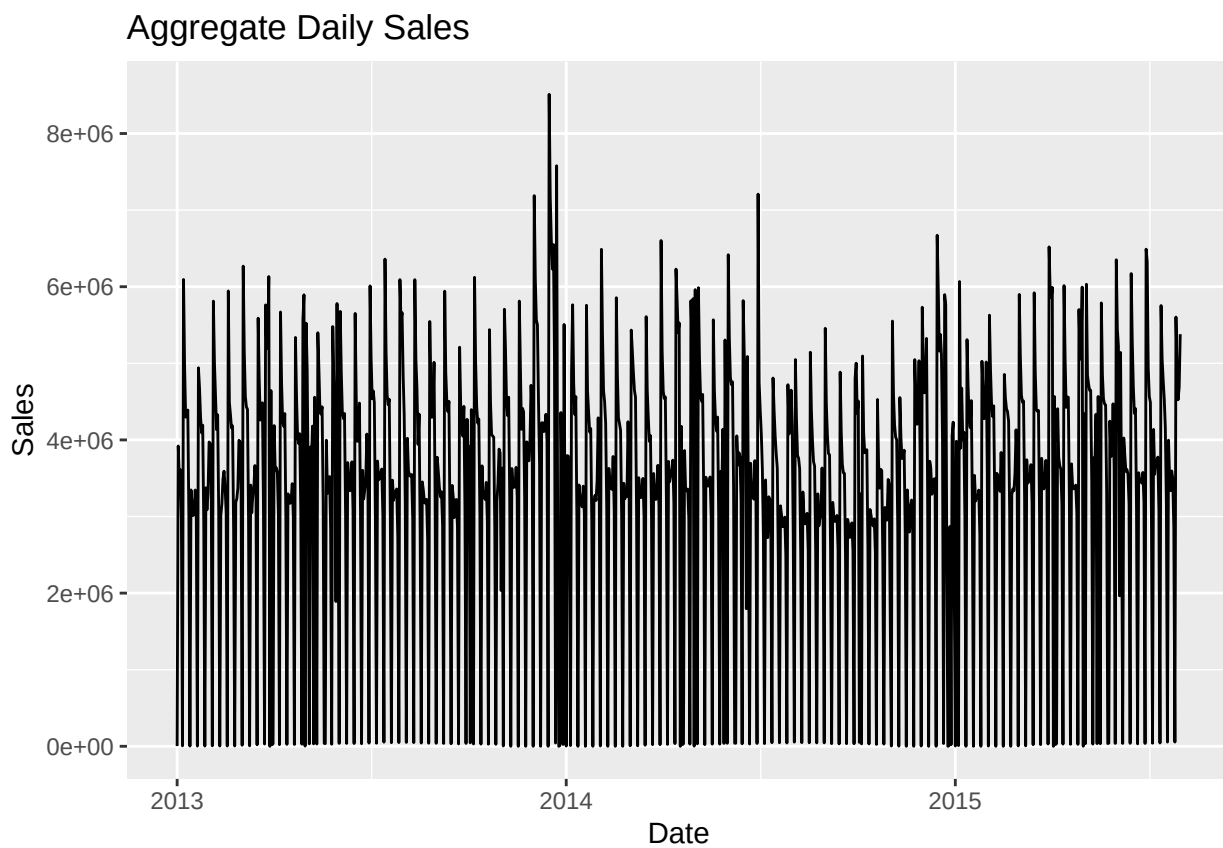
```r
  n_holdout = n_total - n_train,
  train_start = min(train_subset$Date),
  train_end = max(train_subset$Date),
  holdout_start = min(holdout_subset$Date),
  holdout_end = max(holdout_subset$Date)
)

holdout_split
```

```
## # A tibble: 1 x 7
##   n_total n_train n_holdout train_start train_end  holdout_start holdout_end
##     <int>   <dbl>     <dbl> <date>      <date>     <date>        <date>
## 1     942     753       189 2013-01-01  2015-01-23 2015-01-24    2015-07-31
```

Interpretation

To evaluate forecasting performance fairly, we use an 80/20 chronological split: the first 80% of observations are used for model training, and the final 20% are reserved as holdout data. The ETS model is fit only on the training subset, and its forecasts are then compared with actual sales in the holdout period. This allows us to assess out-of-sample forecasting performance under the same split strategy used in the other analyses.

```r
ggplot(daily_data, aes(x = Date, y = Sales)) +
  geom_line() +
  labs(
    title = "Aggregate Daily Sales",
    x = "Date",
    y = "Sales"
  )
```



Interpretation

The aggregate daily sales series shows clear repeated fluctuations over time. In particular, a strong weekly pattern is visible, with regular sharp declines on certain days, likely reflecting weekends or days with very few stores open. This suggests that seasonality is an important feature of the data and supports the use of forecasting methods that can account for recurring weekly structure.

```
sales_ts <- ts(train_subset$Sales, frequency = 7)
```

```
ets_model <- ets(sales_ts)

ets_model
```

```
## ETS(A,N,A)
##
## Call:
## ets(y = sales_ts)
##
##   Smoothing parameters:
##     alpha = 0.0163
##     gamma = 1e-04
##
##   Initial states:
##     l = 3187345.7565
##     s = 1313291 -3304768 -225206.8 575589.6 472373.6 492830.3
##            675890.3
##
##   sigma:  1005260
```

```
##
##      AIC     AICc      BIC
## 25812.93 25813.22 25859.17
```

```r
summary(ets_model)
```

```
## ETS(A,N,A)
##
## Call:
## ets(y = sales_ts)
##
##   Smoothing parameters:
##     alpha = 0.0163
##     gamma = 1e-04
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##   Initial states:
##     l = 3187345.7565
##     s = 1313291 -3304768 -225206.8 575589.6 472373.6 492830.3
##           675890.3
##
##   sigma:  1005260
##
##      AIC     AICc      BIC
## 25812.93 25813.22 25859.17
##
## Training set error measures:
##                    ME     RMSE      MAE MPE MAPE      MASE      ACF1
## Training set 2447.773 999234.2 661286.6 NaN  Inf 0.5947823 0.2933426
```

Interpretation

The selected model is **ETS(M,N,M)**, which means the model uses multiplicative errors, no trend, and multiplicative seasonality. This suggests that the aggregate sales series is mainly characterized by strong seasonal variation rather than a persistent long-term trend. The multiplicative seasonal structure is reasonable here because the size of fluctuations appears to depend on the overall sales level, with larger seasonal swings occurring when sales are higher.

```r
ets_holdout_fc <- forecast(ets_model, h = holdout_horizon)
stl_model <- stlf(sales_ts, h = holdout_horizon)

holdout_predictions <- tibble::tibble(
  Date = holdout_subset$Date,
  actual = holdout_subset$Sales,
  predicted = as.numeric(ets_holdout_fc$mean)
)

stl_holdout_predictions <- tibble::tibble(
  Date = holdout_subset$Date,
  actual = holdout_subset$Sales,
  predicted = as.numeric(stl_model$mean)
)

head(holdout_predictions)
```

```
## # A tibble: 6 x 3
##   Date        actual predicted
```

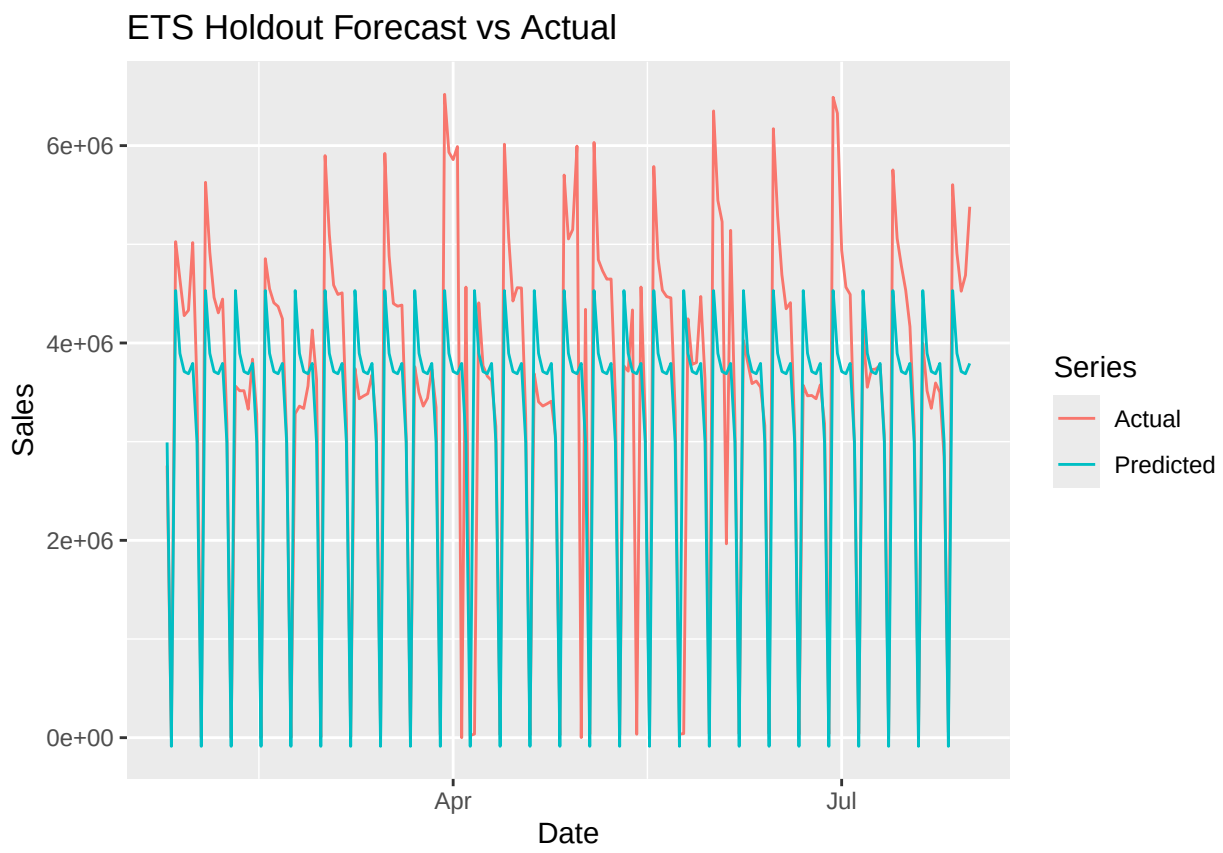
```
##   <date>       <dbl>     <dbl>
## 1 2015-01-24 2757634  2992046.
## 2 2015-01-25       0   -87300.
## 3 2015-01-26 5026610  4530753.
## 4 2015-01-27 4645869  3895021.
## 5 2015-01-28 4277252  3710256.
## 6 2015-01-29 4330827  3688035.
```

```r
rmse_vec <- function(actual, pred) {
  sqrt(mean((actual - pred)^2, na.rm = TRUE))
}

ets_holdout_rmse <- rmse_vec(
  actual = holdout_predictions$actual,
  pred = holdout_predictions$predicted
)

stl_rmse <- rmse_vec(
  actual = holdout_subset$Sales,
  pred = as.numeric(stl_model$mean)
)

ets_model_label <- as.character(ets_model$method)

holdout_rmse_summary <- tibble::tibble(
  model = c(ets_model_label, "STL+ETS"),
  holdout_rmse = c(ets_holdout_rmse, stl_rmse)
)

holdout_rmse_summary
```

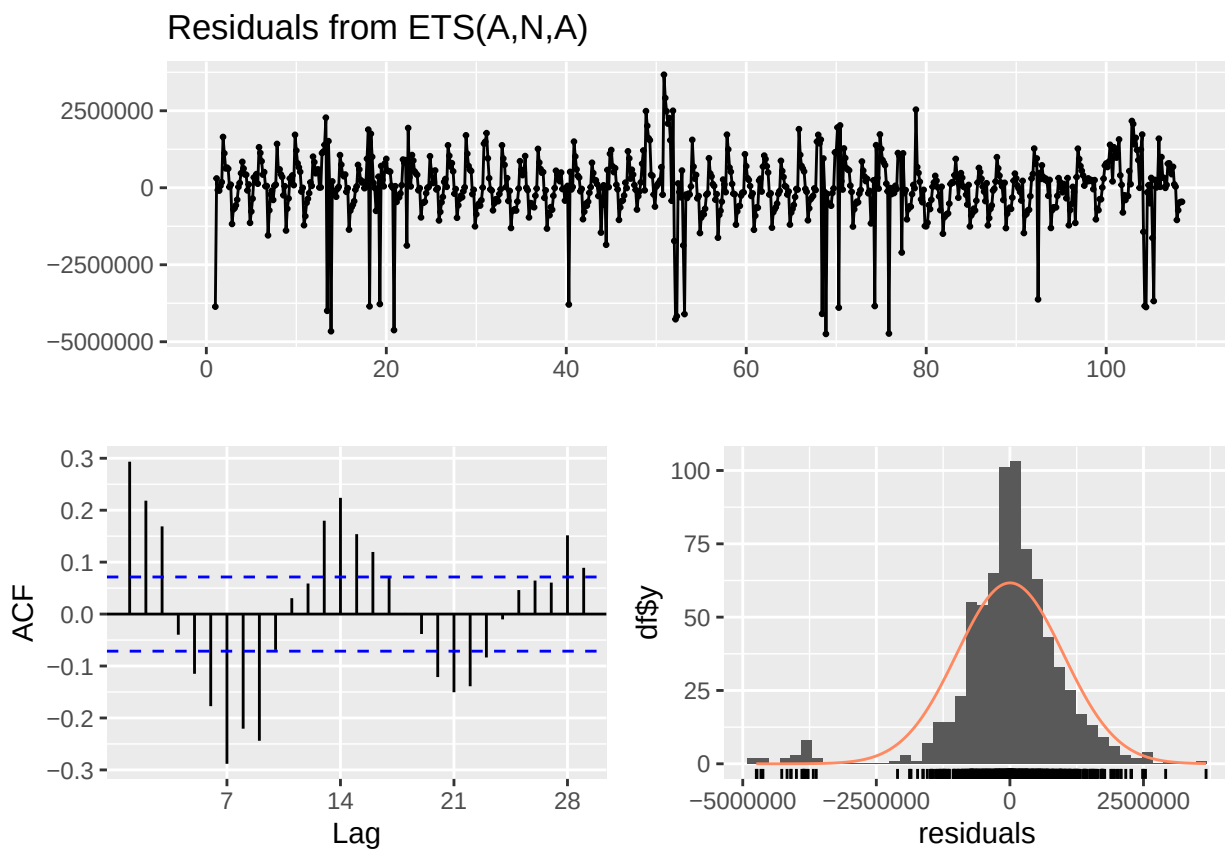
```
## # A tibble: 2 x 2
##   model      holdout_rmse
##   <chr>             <dbl>
## 1 ETS(A,N,A)     1042205.
## 2 STL+ETS        1184604.
```

The RMSE table compares the baseline ETS model and STL+ETS under the same holdout split. In this run, **ETS(A,N,A)** performs better (lower RMSE) than **STL+ETS**. Although residual autocorrelation at seasonal lags motivated testing STL+ETS, the direct ETS specification remains the stronger choice for this dataset and split.

```r
ggplot(holdout_predictions, aes(x = Date)) +
  geom_line(aes(y = actual, color = "Actual")) +
  geom_line(aes(y = predicted, color = "Predicted")) +
  labs(
    title = "ETS Holdout Forecast vs Actual",
    x = "Date",
    y = "Sales",
    color = "Series"
  )
```




```
checkresiduals(ets_model)
```




```
##
##  Ljung-Box test
##
## data:  Residuals from ETS(A,N,A)
## Q* = 374.18, df = 14, p-value < 2.2e-16
##
## Model df: 0.   Total lags used: 14
```

```r
dir.create("results", showWarnings = FALSE, recursive = TRUE)

readr::write_csv(holdout_split, "results/ets_holdout_split.csv")
readr::write_csv(holdout_predictions, "results/ets_holdout_predictions.csv")
readr::write_csv(stl_holdout_predictions, "results/stl_ets_holdout_predictions.csv")
readr::write_csv(holdout_rmse_summary, "results/ets_holdout_rmse_summary.csv")
```